

- A generative agent with fast and slow thinking for complex interactive tasks. *Advances in Neural Information Processing Systems*, 2024, 36
50. Evans J S B, Stanovich K E. Dual-process theories of higher cognition: Advancing the debate. *Perspectives on psychological science*, 2013, 8(3): 223–241
 51. Wang X, Wei J, Schuurmans D, Le Q, Chi E, Narang S, Chowdhery A, Zhou D. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022
 52. Yao S, Yu D, Zhao J, Shafran I, Griffiths T, Cao Y, Narasimhan K. Tree of thoughts: Deliberate problem solving with large language models. *Advances in Neural Information Processing Systems*, 2024, 36
 53. Wang Y, Jiang Z, Chen Z, Yang F, Zhou Y, Cho E, Fan X, Huang X, Lu Y, Yang Y. Recmind: Large language model powered agent for recommendation. *arXiv preprint arXiv:2308.14296*, 2023
 54. Besta M, Blach N, Kubicek A, Gerstenberger R, Gianinazzi L, Gajda J, Lehmann T, Podstawski M, Niewiadomski H, Nyczyk P, others . Graph of thoughts: Solving elaborate problems with large language models. *arXiv preprint arXiv:2308.09687*, 2023
 55. Sel B, Al-Tawaha A, Khattar V, Wang L, Jia R, Jin M. Algorithm of thoughts: Enhancing exploration of ideas in large language models. *arXiv preprint arXiv:2308.10379*, 2023
 56. Gramopadhye M, Szafir D. Generating executable action plans with environmentally-aware language models. In: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2023, 3568–3575
 57. Hao S, Gu Y, Ma H, Hong J J, Wang Z, Wang D Z, Hu Z. Reasoning with language model is planning with world model. *arXiv preprint arXiv:2305.14992*, 2023
 58. Liu B, Jiang Y, Zhang X, Liu Q, Zhang S, Biswas J, Stone P. LLM+P: Empowering large language models with optimal planning proficiency. *arXiv preprint arXiv:2304.11477*, 2023
 59. Dagan G, Keller F, Lascarides A. Dynamic planning with a llm. *arXiv preprint arXiv:2308.06391*, 2023
 60. Yao S, Zhao J, Yu D, Du N, Shafran I, Narasimhan K, Cao Y. React: Synergizing reasoning and acting in language models. In: *The Twelfth International Conference on Learning Representations*. 2023
 61. Song C H, Wu J, Washington C, Sadler B M, Chao W L, Su Y. Llm-planner: Few-shot grounded planning for embodied agents with large language models. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023, 2998–3009
 62. Huang W, Xia F, Xiao T, Chan H, Liang J, Florence P, Zeng A, Tompson J, Mordatch I, Chebotar Y, others . Inner monologue: Embodied reasoning through planning with language models. *arXiv preprint arXiv:2207.05608*, 2022
 63. Madaan A, Tandon N, Gupta P, Hallinan S, Gao L, Wiegrefe S, Alon U, Dziri N, Prabhumoye S, Yang Y, others . Self-refine: Iterative refinement with self-feedback. *Advances in Neural Information Processing Systems*, 2024, 36
 64. Miao N, Teh Y W, Rainforth T. Selfcheck: Using llms to zero-shot check their own step-by-step reasoning. In: *The Twelfth International Conference on Learning Representations*. 2023
 65. Chen P L, Chang C S. Interact: Exploring the potentials of chatgpt as a cooperative agent. *arXiv preprint arXiv:2308.01552*, 2023
 66. Chen Z, Zhou K, Zhang B, Gong Z, Zhao W X, Wen J R. Chatcot: Tool-augmented chain-of-thought reasoning on\chat-based large language models. *arXiv preprint arXiv:2305.14323*, 2023
 67. Nakano R, Hilton J, Balaji S, Wu J, Ouyang L, Kim C, Hesse C, Jain S, Kosaraju V, Saunders W, others . Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*, 2021
 68. Ruan J, Chen Y, Zhang B, Xu Z, Bao T, Du G, Shi S, Mao H, Zeng X, Zhao R. TPTU: Task planning and tool usage of large language model-based AI agents. *arXiv preprint arXiv:2308.03427*, 2023
 69. Patil S G, Zhang T, Wang X, Gonzalez J E. Gorilla: Large language model connected with massive apis. *arXiv preprint arXiv:2305.15334*, 2023
 70. Li M, Song F, Yu B, Yu H, Li Z, Huang F, Li Y. Api-bank: A benchmark for tool-augmented llms. *arXiv preprint arXiv:2304.08244*, 2023
 71. Song Y, Xiong W, Zhu D, Li C, Wang K, Tian Y, Li S. Restgpt: Connecting large language models with real-world applications via restful apis. *arXiv preprint arXiv:2306.06624*, 2023
 72. Liang Y, Wu C, Song T, Wu W, Xia Y, Liu Y, Ou Y, Lu S, Ji L, Mao S, others . Taskmatrix. ai: Completing tasks by connecting foundation models with millions of apis. *Intelligent Computing*, 2024, 3: 0063
 73. Karpas E, Abend O, Belinkov Y, Lenz B, Lieber O, Ratner N, Shoham Y, Bata H, Levine Y, Leyton-Brown K, others . Mrkl systems: A modular, neuro-symbolic architecture that combines large language models, external knowledge sources and discrete reasoning. *arXiv preprint arXiv:2205.00445*, 2022
 74. Ge Y, Hua W, Mei K, Tan J, Xu S, Li Z, Zhang Y, others . Openagi: When llm meets domain experts. *Advances*

- in Neural Information Processing Systems, 2024, 36
75. Surís D, Menon S, Vondrick C. Vipergpt: Visual inference via python execution for reasoning. arXiv preprint arXiv:2303.08128, 2023
76. Bran A M, Cox S, White A D, Schwaller P. Chemcrow: Augmenting large-language models with chemistry tools. arXiv preprint arXiv:2304.05376, 2023
77. Yang Z, Li L, Wang J, Lin K, Azarnasab E, Ahmed F, Liu Z, Liu C, Zeng M, Wang L. Mm-react: Prompting chatgpt for multimodal reasoning and action. arXiv preprint arXiv:2303.11381, 2023
78. Gao C, Lan X, Lu Z, Mao J, Piao J, Wang H, Jin D, Li Y. S3: Social-network simulation system with large language model-empowered agents. arXiv preprint arXiv:2307.14984, 2023
79. Ahn M, Brohan A, Brown N, Chebotar Y, Cortes O, David B, Finn C, Fu C, Gopalakrishnan K, Hausman K, others. Do as i can, not as i say: Grounding language in robotic affordances. arXiv preprint arXiv:2204.01691, 2022
80. Park J S, Popowski L, Cai C, Morris M R, Liang P, Bernstein M S. Social simulacra: Creating populated prototypes for social computing systems. In: Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology. 2022, 1–18
81. Li G, Hammoud H A A K, Itani H, Khizbullin D, Ghanem B. Camel: Communicative agents for "mind" exploration of large scale language model society. arXiv preprint arXiv:2303.17760, 2023
82. al. e T. Auto-GPT. <https://github.com/Significant-Gravitas/Auto-GPT>, 2023
83. Liu R, Yang R, Jia C, Zhang G, Zhou D, Dai A M, Yang D, Vosoughi S. Training socially aligned language models in simulated human society. arXiv preprint arXiv:2305.16960, 2023
84. Chen L, Wang L, Dong H, Du Y, Yan J, Yang F, Li S, Zhao P, Qin S, Rajmohan S, others. Introspective tips: Large language model for in-context decision making. arXiv preprint arXiv:2305.11598, 2023
85. Liu H, Sferrazza C, Abbeel P. Chain of hindsight aligns language models with feedback. In: The Twelfth International Conference on Learning Representations. 2023
86. Yao S, Chen H, Yang J, Narasimhan K. Webshop: Towards scalable real-world web interaction with grounded language agents. Advances in Neural Information Processing Systems, 2022, 35: 20744–20757
87. Dan Y, Lei Z, Gu Y, Li Y, Yin J, Lin J, Ye L, Tie Z, Zhou Y, Wang Y, others. Educhat: A large-scale language model-based chatbot system for intelligent education. arXiv preprint arXiv:2308.02773, 2023
88. Deng X, Gu Y, Zheng B, Chen S, Stevens S, Wang B, Sun H, Su Y. Mind2web: Towards a generalist agent for the web. Advances in Neural Information Processing Systems, 2024, 36
89. Sun R, Arik S O, Nakhosht H, Dai H, Sinha R, Yin P, Pfister T. Sql-palm: Improved large language model adaptation for text-to-sql. arXiv preprint arXiv:2306.00739, 2023
90. Yao W, Heinecke S, Niebles J C, Liu Z, Feng Y, Xue L, Murthy R, Chen Z, Zhang J, Arpit D, Xu R, Mui P, Wang H, Xiong C, Savarese S. Retroformer: Retrospective large language agents with policy gradient optimization, 2023
91. Shu Y, Gu H, Zhang P, Zhang H, Lu T, Li D, Gu N. Rah! recsys-assistant-human: A human-central recommendation framework with large language models. arXiv preprint arXiv:2308.09904, 2023
92. Mandi Z, Jain S, Song S. Roco: Dialectic multi-robot collaboration with large language models. arXiv preprint arXiv:2307.04738, 2023
93. Zhang C, Liu L, Wang J, Wang C, Sun X, Wang H, Cai M. Prefer: Prompt ensemble learning via feedback-reflect-refine. arXiv preprint arXiv:2308.12033, 2023
94. Du Y, Li S, Torralba A, Tenenbaum J B, Mordatch I. Improving factuality and reasoning in language models through multiagent debate. arXiv preprint arXiv:2305.14325, 2023
95. Yang Z, Liu J, Han Y, Chen X, Huang Z, Fu B, Yu G. Appagent: Multimodal agents as smartphone users. arXiv preprint arXiv:2312.13771, 2023
96. Madaan A, Tandon N, Clark P, Yang Y. Memory-assisted prompt editing to improve GPT-3 after deployment. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing. 2022
97. Colas C, Teodorescu L, Oudeyer P Y, Yuan X, Côté M A. Augmenting autotelic agents with large language models. arXiv preprint arXiv:2305.12487, 2023
98. Nascimento N, Alencar P, Cowan D. Self-adaptive large language model (llm)-based multiagent systems. In: 2023 IEEE International Conference on Autonomic Computing and Self-Organizing Systems Companion (ACSOS-C). 2023, 104–109
99. Saha S, Hase P, Bansal M. Can language models teach weaker agents? teacher explanations improve students via theory of mind. arXiv preprint arXiv:2306.09299, 2023
100. Zhuge M, Liu H, Faccio F, Ashley D R, Csordás R, Gopalakrishnan A, Hamdi A, Hammoud H A A K, Herrmann V, Irie K, others. Mindstorms in natural language-based societies of mind. arXiv preprint

- arXiv:2305.17066, 2023
101. Aher G V, Arriaga R I, Kalai A T. Using large language models to simulate multiple humans and replicate human subject studies. In: International Conference on Machine Learning. 2023, 337–371
 102. Akata E, Schulz L, Coda-Forno J, Oh S J, Bethge M, Schulz E. Playing repeated games with large language models. arXiv preprint arXiv:2305.16867, 2023
 103. Ma Z, Mei Y, Su Z. Understanding the benefits and challenges of using large language model-based conversational agents for mental well-being support. In: AMIA Annual Symposium Proceedings. 2023, 1105
 104. Ziems C, Held W, Shaikh O, Chen J, Zhang Z, Yang D. Can large language models transform computational social science? arXiv preprint arXiv:2305.03514, 2023
 105. Horton J J. Large language models as simulated economic agents: What can we learn from homo silicus? Technical report, National Bureau of Economic Research, 2023
 106. Li S, Yang J, Zhao K. Are you in a masquerade? exploring the behavior and impact of large language model driven social bots in online social networks. arXiv preprint arXiv:2307.10337, 2023
 107. Li C, Su X, Fan C, Han H, Xue C, Zheng C. Quantifying the impact of large language models on collective opinion dynamics. arXiv preprint arXiv:2308.03313, 2023
 108. Kovač G, Portelas R, Dominey P F, Oudeyer P Y. The socialai school: Insights from developmental psychology towards artificial socio-cultural agents. arXiv preprint arXiv:2307.07871, 2023
 109. Williams R, Hosseinichimeh N, Majumdar A, Ghafarzadegan N. Epidemic modeling with generative agents. arXiv preprint arXiv:2307.04986, 2023
 110. Jinxin S, Jiabao Z, Yilei W, Xingjiao W, Jiawen L, Liang H. Cgmi: Configurable general multi-agent interaction framework. arXiv preprint arXiv:2308.12503, 2023
 111. Cui J, Li Z, Yan Y, Chen B, Yuan L. Chatlaw: Open-source legal large language model with integrated external knowledge bases. arXiv preprint arXiv:2306.16092, 2023
 112. Hamilton S. Blind judgement: Agent-based supreme court modelling with gpt. arXiv preprint arXiv:2301.05327, 2023
 113. Bail C A. Can generative ai improve social science? 2023
 114. Boiko D A, MacKnight R, Gomes G. Emergent autonomous scientific research capabilities of large language models. arXiv preprint arXiv:2304.05332, 2023
 115. Kang Y, Kim J. Chatmof: An autonomous ai system for predicting and generating metal-organic frameworks. arXiv preprint arXiv:2308.01423, 2023
 116. Swan M, Kido T, Roland E, Santos R P d. Math agents: Computational infrastructure, mathematical embedding, and genomics. arXiv preprint arXiv:2307.02502, 2023
 117. Drori I, Zhang S, Shuttleworth R, Tang L, Lu A, Ke E, Liu K, Chen L, Tran S, Cheng N, others. A neural network solves, explains, and generates university math problems by program synthesis and few-shot learning at human level. Proceedings of the National Academy of Sciences, 2022, 119(32): e2123433119
 118. Chen M, Tworek J, Jun H, Yuan Q, Pinto H P d O, Kaplan J, Edwards H, Burda Y, Joseph N, Brockman G, others. Evaluating large language models trained on code. arXiv preprint arXiv:2107.03374, 2021
 119. Liffiton M, Sheese B E, Savelka J, Denny P. Codehelp: Using large language models with guardrails for scalable support in programming classes. In: Proceedings of the 23rd Koli Calling International Conference on Computing Education Research. 2023, 1–11
 120. Matelsky J K, Parodi F, Liu T, Lange R D, Kording K P. A large language model-assisted education tool to provide feedback on open-ended responses. arXiv preprint arXiv:2308.02439, 2023
 121. Grossmann I, Feinberg M, Parker D C, Christakis N A, Tetlock P E, Cunningham W A. Ai and the transformation of social science research. Science, 2023, 380(6650): 1108–1109
 122. Zhou X, Li G, Liu Z. Llm as dba. arXiv preprint arXiv:2308.05481, 2023
 123. He Z, Wu H, Zhang X, Yao X, Zheng S, Zheng H, Yu B. Chateda: A large language model powered autonomous agent for eda. In: 2023 ACM/IEEE 5th Workshop on Machine Learning for CAD (MLCAD). 2023, 1–6
 124. Huang X, Lian J, Lei Y, Yao J, Lian D, Xie X. Recommender ai agent: Integrating large language models for interactive recommendations. arXiv preprint arXiv:2308.16505, 2023
 125. Deng G, Liu Y, Mayoral-Vilches V, Liu P, Li Y, Xu Y, Zhang T, Liu Y, Pinzger M, Rass S. Pentestgpt: An llm-empowered automatic penetration testing tool. arXiv preprint arXiv:2308.06782, 2023
 126. al. e S. Smolmodels. <https://github.com/smol-ai/developer>, 2023
 127. al. e M U. DemoGPT. <https://github.com/melih-unsal/DemoGPT>, 2023
 128. al. e A O. GPT engineer. <https://github.com/AntonOsika/gpt-engineer>, 2023
 129. Xia Y, Shenoy M, Jazdi N, Weyrich M. Towards autonomous system: flexible modular production system

- enhanced with large language model agents. arXiv preprint arXiv:2304.14721, 2023
130. Ogundare O, Madasu S, Wiggins N. Industrial engineering with large language models: A case study of chatgpt's performance on oil & gas problems. arXiv preprint arXiv:2304.14354, 2023
 131. Zhang C, Yang K, Hu S, Wang Z, Li G, Sun Y, Zhang C, Zhang Z, Liu A, Zhu S C, others . Proagent: Building proactive cooperative ai with large language models. arXiv preprint arXiv:2308.11339, 2023
 132. Hu B, Zhao C, others . Enabling intelligent interactions between an agent and an llm: A reinforcement learning approach. arXiv:2306.03604, 2023
 133. Wu Y, Min S Y, Bisk Y, Salakhutdinov R, Azaria A, Li Y, Mitchell T, Prabhunoye S. Plan, eliminate, and track—language models are good teachers for embodied agents. arXiv preprint arXiv:2305.02412, 2023
 134. Zhang D, Chen L, Zhang S, Xu H, Zhao Z, Yu K. Large language models are semi-parametric reinforcement learning agents. *Advances in Neural Information Processing Systems*, 2024, 36
 135. Di Palo N, Byravan A, Hasenclever L, Wulfmeier M, Heess N, Riedmiller M. Towards a unified agent with foundation models. In: *Workshop on Reincarnating Reinforcement Learning at ICLR 2023*. 2023
 136. Wu J, Antonova R, Kan A, Lepert M, Zeng A, Song S, Bohg J, Rusinkiewicz S, Funkhouser T. Tidybot: Personalized robot assistance with large language models. arXiv preprint arXiv:2305.05658, 2023
 137. Wu Z, Wang Z, Xu X, Lu J, Yan H. Embodied task planning with large language models. arXiv preprint arXiv:2307.01848, 2023
 138. Dasgupta I, Kaeser-Chen C, Marino K, Ahuja A, Babayan S, Hill F, Fergus R. Collaborating with language models for embodied reasoning. arXiv preprint arXiv:2302.00763, 2023
 139. Nottingham K, Ammanabrolu P, Suhr A, Choi Y, Hajishirzi H, Singh S, Fox R. Do embodied agents dream of pixelated sheep?: Embodied decision making using language guided world modelling. In: *Workshop on Reincarnating Reinforcement Learning at ICLR 2023*. 2023
 140. Zhou W, Peng X, Riedl M. Dialogue shaping: Empowering agents through npc interaction. arXiv preprint arXiv:2307.15833, 2023
 141. Li H, Hao Y, Zhai Y, Qian Z. The hitchhiker's guide to program analysis: A journey with large language models. arXiv preprint arXiv:2308.00245, 2023
 142. al. e R. AgentGPT. <https://github.com/reworkd/AgentGPT>, 2023
 143. al. e E. Ai-legion. <https://github.com/eumemic/ai-legion>, 2023
 144. al. e J X. Agixt. <https://github.com/Josh-XT/AGiXT>, 2023
 145. al. e C. Xlang. <https://github.com/xlang-ai/xlang>, 2023
 146. al. e N. Babyagi. <https://github.com/yoheinakajima>, 2023
 147. Chase H. langchain. <https://docs.langchain.com/docs/>, 2023
 148. al. e A M. WorkGPT. <https://github.com/team-openpm/workgpt>, 2023
 149. al. e F R. LoopGPT. <https://github.com/farizrahman4u/loopgpt>, 2023
 150. al. e A E. GPT-researcher. <https://github.com/assafelovic/gpt-researcher>, 2023
 151. Qin Y, Hu S, Lin Y, Chen W, Ding N, Cui G, Zeng Z, Huang Y, Xiao C, Han C, others . Tool learning with foundation models. arXiv preprint arXiv:2304.08354, 2023
 152. Face H. transformers-agent. https://huggingface.co/docs/transformers/transformers_agents, 2023
 153. al. e E. Miniagi. <https://github.com/muellerberndt/mini-agi>, 2023
 154. al. e T. Superagi. <https://github.com/TransformerOptimus/SuperAGI>, 2023
 155. Wu Q, Bansal G, Zhang J, Wu Y, Zhang S, Zhu E, Li B, Jiang L, Zhang X, Wang C. Autogen: Enabling next-gen llm applications via multi-agent conversation framework. arXiv preprint arXiv:2308.08155, 2023
 156. Chen W, Su Y, Zuo J, Yang C, Yuan C, Qian C, Chan C M, Qin Y, Lu Y, Xie R, others . Agentverse: Facilitating multi-agent collaboration and exploring emergent behaviors in agents. arXiv preprint arXiv:2308.10848, 2023
 157. Lee M, Srivastava M, Hardy A, Thickstun J, Durmus E, Paranjape A, Gerard-Ursin I, Li X L, Ladhak F, Rong F, others . Evaluating human-language model interaction. arXiv preprint arXiv:2212.09746, 2022
 158. Chan C M, Chen W, Su Y, Yu J, Xue W, Zhang S, Fu J, Liu Z. Chateval: Towards better llm-based evaluators through multi-agent debate. arXiv preprint arXiv:2308.07201, 2023
 159. Kang S, Yoon J, Yoo S. Large language models are few-shot testers: Exploring llm-based general bug reproduction. In: *2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE)*. 2023, 2312–2323
 160. Jalil S, Rafi S, LaToza T D, Moran K, Lam W. Chatgpt and software testing education: Promises & perils. In: *2023 IEEE International Conference on Software Test-*

- ing, Verification and Validation Workshops (ICSTW). 2023, 4130–4137
161. Mehta N, Teruel M, Sanz P F, Deng X, Awadallah A H, Kiseleva J. Improving grounded language understanding in a collaborative environment by interacting with agents through help feedback. *arXiv preprint arXiv:2304.10750*, 2023
 162. Chen A, Phang J, Parrish A, Padmakumar V, Zhao C, Bowman S R, Cho K. Two failures of self-consistency in the multi-step reasoning of llms. *arXiv preprint arXiv:2305.14279*, 2023
 163. Choi M, Pei J, Kumar S, Shu C, Jurgens D. Do llms understand social knowledge? evaluating the sociability of large language models with socket benchmark. *arXiv preprint arXiv:2305.14938*, 2023
 164. Zhang D, Chen L, Zhao Z, Cao R, Yu K. Mobile-env: An evaluation platform and benchmark for interactive agents in llm era. *arXiv preprint arXiv:2305.08144*, 2023
 165. Chalamalasetti K, Götze J, Hakimov S, Madureira B, Sadler P, Schlangen D. clembench: Using game play to evaluate chat-optimized language models as conversational agents. *arXiv preprint arXiv:2305.13455*, 2023
 166. Lin J, Tomlin N, Andreas J, Eisner J. Decision-oriented dialogue for human-ai collaboration. *arXiv preprint arXiv:2305.20076*, 2023
 167. Feldt R, Kang S, Yoon J, Yoo S. Towards autonomous testing agents via conversational large language models. *arXiv preprint arXiv:2306.05152*, 2023
 168. Liang Y, Zhu L, Yang Y. Tachikuma: Understanding complex interactions with multi-character and novel objects by large language models. *arXiv preprint arXiv:2307.12573*, 2023
 169. Liu X, Yu H, Zhang H, Xu Y, Lei X, Lai H, Gu Y, Ding H, Men K, Yang K, others . Agentbench: Evaluating llms as agents. *arXiv preprint arXiv:2308.03688*, 2023
 170. Liu Z, Yao W, Zhang J, Xue L, Heinecke S, Murthy R, Feng Y, Chen Z, Niebles J C, Arpit D, others . Bolla: Benchmarking and orchestrating llm-augmented autonomous agents. *arXiv preprint arXiv:2308.05960*, 2023
 171. Xu B, Liu X, Shen H, Han Z, Li Y, Yue M, Peng Z, Liu Y, Yao Z, Xu D. Gentopia. ai: A collaborative platform for tool-augmented llms. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*. 2023, 237–245
 172. Huang J t, Lam M H, Li E J, Ren S, Wang W, Jiao W, Tu Z, Lyu M R. Emotionally numb or empathetic? evaluating how llms feel using emotionbench. *arXiv preprint arXiv:2308.03656*, 2023
 173. Zhou S, Xu F F, Zhu H, Zhou X, Lo R, Sridhar A, Cheng X, Bisk Y, Fried D, Alon U, others . Webarena: A realistic web environment for building autonomous agents. *arXiv preprint arXiv:2307.13854*, 2023
 174. Banerjee D, Singh P, Avadhanam A, Srivastava S. Benchmarking llm powered chatbots: methods and metrics. *arXiv preprint arXiv:2308.04624*, 2023
 175. Zhao W X, Zhou K, Li J, Tang T, Wang X, Hou Y, Min Y, Zhang B, Zhang J, Dong Z, others . A survey of large language models. *arXiv preprint arXiv:2303.18223*, 2023
 176. Yang J, Jin H, Tang R, Han X, Feng Q, Jiang H, Zhong S, Yin B, Hu X. Harnessing the power of llms in practice: A survey on chatgpt and beyond. *ACM Transactions on Knowledge Discovery from Data*, 2023
 177. Wang Y, Zhong W, Li L, Mi F, Zeng X, Huang W, Shang L, Jiang X, Liu Q. Aligning large language models with human: A survey. *arXiv preprint arXiv:2307.12966*, 2023
 178. Huang J, Chang K C C. Towards reasoning in large language models: A survey. *arXiv preprint arXiv:2212.10403*, 2022
 179. Mialon G, Dessì R, Lomeli M, Nalmpantis C, Pasunuru R, Raileanu R, Rozière B, Schick T, Dwivedi-Yu J, Celikyilmaz A, others . Augmented language models: a survey. *arXiv preprint arXiv:2302.07842*, 2023
 180. Chang Y, Wang X, Wang J, Wu Y, Yang L, Zhu K, Chen H, Yi X, Wang C, Wang Y, others . A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*, 2023
 181. Chang T A, Bergen B K. Language model behavior: A comprehensive survey. *Computational Linguistics*, 2024, 1–58
 182. Li C, Wang J, Zhu K, Zhang Y, Hou W, Lian J, Xie X. Emotionprompt: Leveraging psychology for large language models enhancement via emotional stimulus. *arXiv e-prints*, 2023, arXiv–2307
 183. Zhuo T Y, Li Z, Huang Y, Shiri F, Wang W, Haffari G, Li Y F. On robustness of prompt-based semantic parsing with large pre-trained language model: An empirical study on codex. *arXiv preprint arXiv:2301.12868*, 2023
 184. Gekhman Z, Oved N, Keller O, Szpektor I, Reichart R. On the robustness of dialogue history representation in conversational question answering: a comprehensive study and a new prompt-based method. *Transactions of the Association for Computational Linguistics*, 2023, 11: 351–366
 185. Ji Z, Lee N, Frieske R, Yu T, Su D, Xu Y, Ishii E, Bang Y J, Madotto A, Fung P. Survey of hallucination in